Intelligent Document Processing System

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Abstract

This study proposes and prototypes an Intelligent Document Processing (IDP) system designed to automate the classification and complex analysis of diverse, real-world documents. The system uses a subset of RVL-CDIP dataset for efficient simulation and rapid architectural validation. The IDP system is a hybridized multimodal pipeline that integrates visual and textual intelligence. Classification is achieved through parallel pathways: a CNN module for document layout and feature and OCR/LSTM module that processes the extracted text content. The core innovation lies on the Fuzzy logic module, which acts as a robust decision layer by integrating confidence scores from both CNN and LSTM predictions to output to a final certainty confidence. The pipeline also uses BART transformer for abstractive summarization and the mBART-50 transformer for multilingual environments. The successful fusion of deep learning (CNN/LSTM/Transformers) with Fuzzy Logic establishes a crucial foundation for deploying trustworthy, high-throughput IDP solutions in enterprise environments.

Keywords
Intelligent Document Processing (IDP), Convolution Neural Network (CNN), Long Short-Term Memory (LSTM), Fuzzy Logic, Document Classification, Multimodal Learning, Summarization, Translation.

1 Introduction

The growing digitization of business processes has led to a massive amount of unstructured and semi-structured documents such as bills, resumes, questionnaires, agreements, memos, budgets, and scanned documents. Businesses dealing with finance, HR, health, education, and legal services work with thousands of such documents, which are converted into PDF files and scanned documents. These operations are not scalable and are time, labor, and resource-intensive, which has led to the increased adoption of Intelligent Document Processing System (IDPS) [8], [10], [13].

Building such a robust IDPS, however, involves a number of real-world challenges. Real-world documents exhibit considerable variability with respect to layout, font, resolution, orientation, noise, illumination, and text density. Various document classes also often share common visual and semantic attributes, such as resumes resembling questionnaires and memos resembling structured documents. These problems are further aggravated when optical character recognition (OCR) output is impaired by poor scans and handwritten information [13], [16], [17]. Resolving such problems and incorporating a common solution to IDPS design and implementation require a joint solution involving visual and textual information accompanied by management of ambiguities during the decision-making stage [8], [10], [14].

The standard test data used to classify documents is the RVL-CDIP (Ryerson Vision Lab Complex Document Information Processing) set, which is a set of 400,000 grayscale images, equally balanced into 16 classes [11], [16], [17]. The total set is comprised of 320,000 training samples, 40,000 evaluation samples, and 40,000 testing samples, each resized such that the longest side is not more than 1000 pixels. RVL-CDIP, while very robust and having high diversity, is not suitable for training large deep models because it is very time and resource-intensive and occupies considerable GPU RAM. In view of fast prototyping using a computer having limited resources, a light variant of RVL-CDIP is used.

The design of the Intelligent Document Processing System is conducted with the help of

the PyTorch tool and involves a multimodal, multi-stage architecture. The system has two major paths to document processing: Path 1 is the CNN-based image categorization part, where document images go through preprocessing, resizing, and then are fed into a convolutional neural network to obtain structural information such as document layout, text, tables, and margins [15], [16], [17]. Path 2 is the OCR-based Text Understanding Path, which involves the extracted text being subjected to classification using the LSTM/BiLSTM classifier [3], [7], [15], and another part is the BART Transformer Model, which is able to perform summarization and translation tasks [1], [2], [19]. The document analysis system also has a fuzzy logic decision part to resolve possible discrepancies between image and text models by effectively combining the output and obtaining an integrated result, such as "Letter - 98 percent confidence" [4], [12], [18]. The proposed system is a unified architecture involving image perception, modeling text sequences, soft computing, and transformers. With the synergy of CNNs, OCR, LSTM Encoder, fuzzy classification, and advanced transformers, the proposed IDPS system brings forth an efficient and flexible solution to the document classification, summarization, and translation tasks, which are remarkably aligned to the needs of businesses in the current market [20].

2 Literature Review

Intelligent document processing has advanced rapidly due to the evolution of deep neural architectures capable of modeling visual, textual, and semantic patterns. Initial progress was achieved when Sutskever et al. introduced the sequence-to-sequence (Seq2Seq) framework using encoder-decoder recurrent neural networks, demonstrating the ability of LSTM models to learn long-range dependencies and generate coherent text sequencing [3]. This architecture had become a cornerstone for Optical Character Recognition (OCR) post-processing, document text reconstruction, and sequential classification within document analysis systems. Transformer architectures revolutionized language modeling by replacing recurrence with self-attention mechanisms. Liu et al. further extended this model by designing very deep transformer networks for neural machine translation, addressing instability and achieving contextual representations essential for downstream document understanding tasks [1]. These deep contextual embeddings are particularly beneficial in scenarios where documents contain complex structures and variable layouts.

The progress in abstractive summarization also empowered modern intelligent document processing pipelines. A new gap-sentence generation objective was introduced by the PEGASUS model by Zhang et al., enabling the model to learn summary-like representations during pre-training and increasing performance especially on long-form document summarization tasks [2]. In tandem, structured information was able to be extracted with high accuracy from documents thanks to strong Named Entity Recognition architectures. Lample et al. proposed a hybrid model including bidirectional LSTMs with CRFs and character-level embeddings, achieving state-of-the-art performance in entity extraction tasks by capturing both local morphological features and global contextual dependencies effectively [7]. These NER techniques directly support automated processing for invoices, forms, contracts, and resumes, where the correct identification of names, dates, amounts, and labels is crucial.

While deep learning yields good performance regarding modeling capability, the real-world document images usually exhibit noise, blur, or inconsistencies that introduce ambiguity in classification and OCR outputs. Consequently, there has been great interest in using fuzzy logic as an uncertainty-handling mechanism to address the challenges. Nedeljkovic showed the effectiveness of fuzzy rule-based reasoning for image classification. It demonstrated that fuzzy membership functions enhance robustness in cases where visual boundaries or image features are not well defined [4]. In addition, texture-oriented methods such as the Gray-Level Co-Occurrence Matrix (GLCM)-related approaches, investigated by Iqbal et al. on crop classification, underlined the importance of structural feature extraction methods, which can be exploited in layout-based document analysis [5]. Finally, the key contribution of Hossin and Sulaiman was related to the proper selection of performance metrics and evaluation strategies of the classification system. This ensures the reliable assessment of the hybrid models combined from CNNs, RNNs, transformers, and fuzzy inference modules [6]. Conclusively, these studies provide a comprehensive basis for the development of robust intelligent document-processing systems that incorporate CNN-based OCR, LSTM-driven sequence modeling, transformer-based contextual understanding, NER extraction, and fuzzy logic for uncertainty management.

3 Simulation Analysis

3.1 Dataset Description

This task utilizes a Ryerson Vision Lab CDIP document image classification benchmark, known as RVL-CDIP, from the RVL-CDIP dataset. The original RVL-CDIP contains:

- 400,000 grayscale document images.
- 16 document types ('advertisement', 'budget', 'email', 'file folder', 'form', 'handwritten', 'invoice', 'letter', 'memo', 'news article', 'presentation', 'questionnaire', 'resume', 'scientific publication', 'scientific report', 'specification').
- About 25,000 images for each class.
- 320,000 training images.
- 40,000 validation images.
- 40,000 test images.

However, the entire data requires a considerable amount of computation. Training deep nets on 400k images requires:

- High-end GPUs that require multiple or larger memory.
- Lengthy training periods, from hours to days - enough storage space memory.
- Considerable preprocessing overhead Based on these constraints, and to provide a quick prototype, our system does not make use of the entire RVL-CDIP.

. Instead, a derived test subset of RVL-CDIP is used that retains exemplar instances from multiple classes but is of a size that makes possible:

- Quicker experimentation.
- Rapid prototyping.
- Training and inference on modest hardware.
- Rapid iteration for model development and preprocessing.

It is able to retain document variability (noise, skew, varying font sizes, differences in layouts), thus keeping the system realistic even when less computation is required.

3.1.1 How the Dataset Fits the Pipelines

It relates to two processing pipes. (a) CNN Classification Pipeline - Images are pre-processed and resized and only visual features are used. Ground truth labels train the CNN classifier. (b) OCR Text Understanding Pipeline - Original images aren't distorted by preprocessing. It extracts the text content. Extracted text feeds into: LSTM for text-based document classification. (c) BART Transformer for Summarization and Translation - By using the same data for both pathways, consistency is maintained, and multimodal learning is possible.

3.1.2 Why a Reduced Dataset Still Suffices

Even though a subset is being utilized, this is good enough because:

- The purpose is simulation and analysis, not bulk deployment.
- The pipeline architecture, although not full accuracy benchmarking, is of importance.
- hmarking, is of importance.
- The subset nevertheless provides varied instances for a just evaluation of:
  - OCR performance
  - CNN classification
  - LSTM-based text analysis
  - Fuzzy logic confidence scoring
  - Transformer-based summarization and translation.

In summary, this reduced data set remains representative as all experiments can be performed effectively under these constraints.

3.2 Image Preprocessing

Image preprocessing is done in only for CNN in this intelligent document processing system. The image preprocessing functions includes transforms such as resize, grey scaling, random affine, random rotation and normalization of the image. One function specifically used for CNN is resizing. The images are resized to 512x512 pixels.

3.3 Model Architecture

The model's architecture as seen in the above image comprises of a CNN module, an OCR module, a LSTM module, a fuzzy logic module, a summarizer and a translator. The libraries and

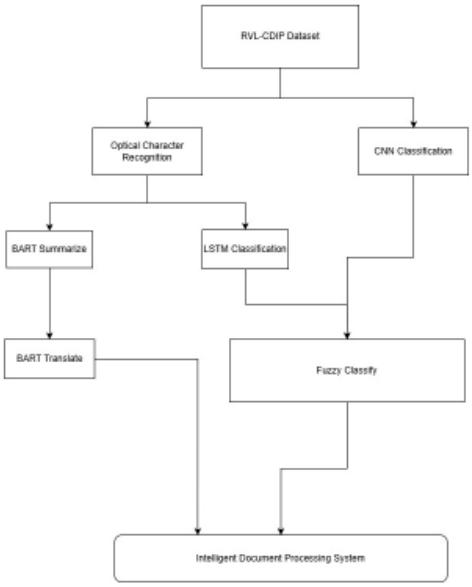


Figure 1: Complete Model Architecture

pretrained models used are: OpenCV, Pillow, scikit-fuzzy networks, pytesseract, transformers, accelerate, tiktoken, sentencepiece, NumPy version 1.26.4. The main framework around which this model is built is PyTorch.

3.3.1 CNN Module

The model uses a basic 2-dimensional CNN module with 4 convolution blocks, each block consists of a 2-dimensional convolution layer with input channels, kernel size and paddings predefined, a 2-dimensional batch normalization block, a max poling layer with kernel size and stride clearly defined and a value defined dropout layer. As stated earlier, the CNN module has its own preprocessing module for the images. The training and testing split are done in 80:20 ratio. The training batch loader is set with a batch size of 64 with shuffle set to true as well. The same logic is followed for the validation set loader as well. CNN once trained generates a best model.pth which is then used in the complete pipelining.

3.3.2 OCR Module

Optical character recognition is used in this model to generate text from the given images. This OCR module uses pytesseract which is a predefined model for text generation. All the image files in the dataset are passed through a function to convert numpy array to PIL image and then using the pytesseract model, the image is converted to text. These texts are then loaded into a CSV file along with their respective labels. The CSV file is later utilized in the LSTM module for classification.

3.3.3 LSTM Module

The LSTM classification module used in this model takes its input from the text generated from the OCR module which is stored in the CSV file. LSTM makes use of a simple text tokenizer, which is a simple word level tokenizer which helps to build vocabulary from lists of texts. The training and validation split follows the 80:20 train test ratio. The module generates weights in the form of best model.pt and fine-LSTM model.pt which are then used in the complete pipelining.

3.3.4 Fuzzy Logic

The Fuzzy inference module used in this model combines the CNN and LSTM predictions. It creates a fuzzy logic system with the CNN and LSTM inputs it receives in the form of best model.pt. The output for the fuzzy module is in terms of confidence for both CNN and LSTM models as, “Low, Medium, or High”. The final output of the Fuzzy module is in terms of certainty confidences, “Uncertain, Moderate or Certain”. These confidences are based of rules which are stated as follows:

```
rule1 = ctrl.Rule(cnn_conf['high'] & lstm_conf['
     high'],
                certainty['certain'])

rule2 = ctrl.Rule(cnn_conf['high'] & lstm_conf['
     medium'],
                certainty['moderate'])

rule3 = ctrl.Rule(cnn_conf['medium'] & lstm_conf[
     'high'],
                certainty['moderate'])

rule4 = ctrl.Rule(cnn_conf['medium'] & lstm_conf[
     'medium'],
                certainty['moderate'])

rule5 = ctrl.Rule(cnn_conf['low'] & lstm_conf['
     low'],
                certainty['uncertain'])

rule6 = ctrl.Rule(cnn_conf['high'] & lstm_conf['
     low'],
                certainty['moderate'])

rule7 = ctrl.Rule(cnn_conf['low'] & lstm_conf['
     high'],
                certainty['moderate'])
```

3.3.5 Summarization

The summarizer uses inbuilt models from transformers such as BartTokenizer and BartForConditionalGeneration. Once executed, the module can be used to summarize text by loading them from a local folder.

3.3.6 Translation

The translator uses MBartForConditionalGeneration and MBart50Tokenizer. The translation can be done in 50 different languages by entering the code for the respective language.

3.3.7 Complete Pipelining

The pipelining is an integrated module combining all the previously defined modules. The inputs to the pipeline are the paths to trained CNN and LSTM weights which were earlier stored in best$_{m}odel.pt$.

3.3.8 Path Simulation

Once CNN, LSTM, Fuzzy, Summarization and Translation modules are loaded, the pipeline is initialized, and the image path is fed into the pipeline. The CNN classification is done first and parallelly the text is extracted from OCR. With the text extracted from OCR, LSTM classification is proceeded with. Fuzzy logic classification starts with integrating the CNN and LSTM pipelines into the module and its classification report is generated. The texted extracted from OCR is in turn used to summarize and translate the respective documents. A simulation of the path followed by the image as a raw .tif file to being processed by the model is depicted below:

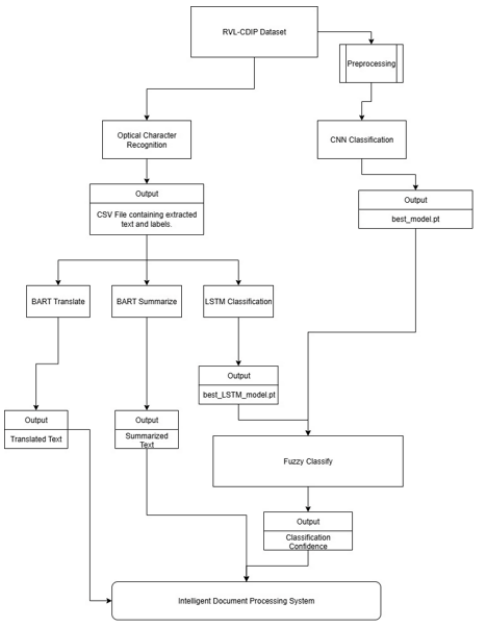


Figure 2: Complete Simulation Path

4 Results and Discussion

A subset of the RVL-CDIP dataset containing 40,000 samples is used in this model due to the memory and GPU constraints. A training set of 32,000 samples and a validation set of 8,000 is used as the 80:20 train test split.

4.1 Individual Models

4.1.1 CNN Results

CNN training is done with a resolution of 512x512 pixels and a batch size of 50 for the 40,000 samples which states that 800 batches of training data is used to train the model. Cross entropy loss and Adam's optimizer is used as the loss function and the optimizer. The model weights once trained will be saved as output in best model.pth.The training accuracy and validation accuracy of the model is shown in the image below:

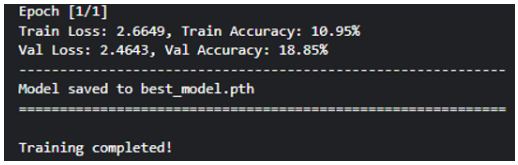


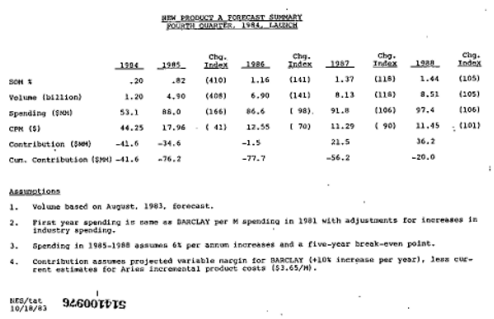
Figure 3: CNN Result

As seen, training and validation accuracy of 10.95 percentage and 18.85 percentage with a very high loss estimate of 2.66 and 2.46 percentage respectively makes us conclude that CNN is not the ideal model for the RVL-CDIP dataset.

4.1.2 OCR Results

To generate individual results for the optical character recognition, an individual file is loaded in the image path. With the help of pytesseract, an existing OCR model, image is converted to string. Example image path: "/kaggle/input/the-rvlcdip-dataset-test/test/budget/0000011677.tif"

The raw image in the dataset:



NEW PRODUCT A FORECAST SUMMARY
FOURTH QUARTER, 1984, FAGERCH

	1984	1985	Chg. Index	1986	Chg. Index	1987	Chg. Index	1988	Chg. Index
SOM %	.20	.82	(410)	1.16	(141)	1.37	(118)	1.44	(105)
Volume (billion)	1.20	4.90	(408)	6.90	(141)	8.13	(118)	8.51	(105)
Spending ($MM)	53.1	88.0	(166)	86.6	(98)	91.8	(106)	97.4	(106)
CPM ($)	44.25	17.96	(41)	12.55	(70)	11.29	(90)	11.45	(101)
Contribution ($MM)	-41.6	-34.6		-1.5		21.5		36.2	
Cum. Contribution ($MM)	-41.6	-76.2		-77.7		-56.2		-20.0	

Assumptions

1. Volume based on August, 1983, forecast.
2. First year spending is same as BARCLAY per M spending in 1981 with adjustments for increases in industry spending.
3. Spending in 1985-1988 assumes 6% per annum increases and a five-year break-even point.
4. Contribution assumes projected variable margin for BARCLAY (+10% increase per year), less current estimates for Arias incremental product costs ($2.65/M).

RFS/tat
10/18/83

Figure 4: Raw image

The extracted Text:

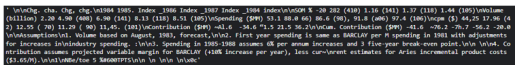


Figure 5: Extracted Image

Moreover, to reduce the models time complexity, all 40,000 image samples were converted to text and stored in a CSV file as a one-time process, thereby, making a novel dataset of our own which will be further used in the LSTM model and in the summarization and translation modules.

4.1.3 LSTM Results

Based on the OCR text for all the 40,000 samples consolidated into one CSV file, LSTM classification training is performed. The LSTM Training is done for a total of 20 epochs. The training and validation accuracy along with the training and validation loss can be seen in the image below:

```
Epoch [10/20]
  Train Loss: 0.2921 | Train Acc: 0.9139
  Val Loss:   1.2789 | Val Acc:   0.7191
  ⚠ No improvement (5/5)
------------------------------------------------------------
Early stopping at epoch 10
Training completed!
```

Figure 6: Training and Validation Accuracy

Early stopping is introduced as well when no improvements are seen in the validation loss function. The Evaluation Results are as seen in the image below:

```
============================================================
EVALUATION RESULTS
============================================================
Overall Accuracy: 0.7074

Classification Report:
              precision    recall  f1-score   support

           0       0.58      0.46      0.52       503
           1       0.70      0.69      0.69       501
           2       0.88      0.87      0.87       503
           3       0.47      0.72      0.57       506
           4       0.65      0.64      0.64       501
           5       0.50      0.50      0.50       507
           6       0.78      0.74      0.76       495
           7       0.76      0.72      0.74       493
           8       0.78      0.79      0.78       498
           9       0.71      0.57      0.63       493
          10       0.58      0.59      0.59       498
          11       0.76      0.78      0.77       487
          12       0.95      0.97      0.96       507
          13       0.83      0.80      0.81       514
          14       0.62      0.67      0.65       500
          15       0.90      0.82      0.86       494

    accuracy                           0.71      8000
   macro avg       0.72      0.71      0.71      8000
weighted avg       0.72      0.71      0.71      8000
```

Figure 7: Evaluation Metric

The plot for training and validation loss and accuracy:

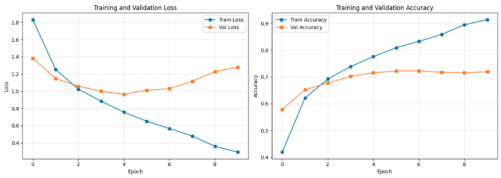


Figure 8: Loss and Accuracy

We can conclude that LSTM is a better fit model for classification of the RVL-CDIP dataset compared to CNN.

4.1.4 Complete Pipeline Results

Once the complete pipeline is initialized, the best model paths which is nothing but the weights of the CNN and LSTM models are loaded in as arguments to the Intelligent Document Processor class and the required output is generated which comprises of the CNN classification, the OCR text extraction, the LSTM Classification, The fuzzy prediction, the summary of the extracted text and the translation of the summary. Example: memo/00040465.tif' (the image to be processed on)

Initializing of the Pipeline:

```
============================================================
INITIALIZING DOCUMENT PROCESSING PIPELINE
============================================================

[1/6] Loading CNN classifier...
✓ CNN classifier ready

[2/6] Loading LSTM classifier...
✓ LSTM classifier ready

[3/6] Initializing fuzzy classification system...
✓ Fuzzy system ready

[4/6] Loading BART summarization model...
✓ Summarizer ready

[5/6] Loading mBART translation model...
✓ Translator ready

[6/6] Pipeline initialization complete!
============================================================
```

Figure 9: Initializing of the Pipeline

Processing Document:

```
[STEP 1/7] Preprocessing image...
✓ Image preprocessed for CNN

[STEP 2/7] Running CNN classification...
✓ CNN classification complete: tensor([[-0.1886,  0.0154, -0.1404, -0.0387,  0.0023, -0.1124, -0.0234,  0.0009,
          0.0373, -0.0451,  0.0283, -0.0252,  0.0639,  0.0690,  0.0055, -0.0141]],
        device='cuda:0')

[STEP 3/7] Extracting text with OCR...
✓ Extracted 450 characters

[STEP 4/7] Running LSTM text classification...
✓ LSTM classification complete: tensor([[-5.5061e-01,  9.3061e-01,  8.6275e-01,  8.8835e-01,  2.2403e-01,
         -4.2738e-01, -3.7125e-01, -1.4039e+00, -1.1951e+00, -4.8847e-01,
          3.2263e+00, -8.4442e-01, -2.4512e+00, -1.9972e+00,  8.8925e-01,
         -3.1910e-03]], device='cuda:0')

[STEP 5/7] Applying fuzzy logic classification...
✓ Fuzzy classification complete: presentation

[STEP 6/7] Generating summary...
```

Figure 10: Processing Document 1

Figure 11: Processing Document 2

A complete step by step processing of the document is seen in the images above. The accuracy of the CNN model affects the overall prediction, whereas the extracted text from OCR is successfully summarized and translated. Future work would be to work on improving the accuracy of the CNN model so as to get a better prediction from the Fuzzy Classification.

5 Conclusion

The project establishes a robust and technically validated Intelligent Document Processing (IDP) system. The document classification is made reliable by combining two distinct prediction pathways, the CNN classification pipeline which analyzes the document's visual layout and the OCR and text understanding pipeline, which extracts text using pytesseract to give input to the LSTM classification module. The two distinct pipeline merges at the fuzzy module. This is the critical component because it gives the final certanity confidence. The fuzzy module uses seven specific rules for managing ambiguity and enhancing classification reliability. The BART summarizer condenses document test and then it can be converted into 50 different languages using mBART translator, this makes the IDP high volume enterprise application.

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